CSM
Spring 2026

EECS 16B
Worksheet 5

1. Phasor Analysis

Any sinusoidal time-varying function $v(t)$, representing a voltage or a current, can be expressed in the form

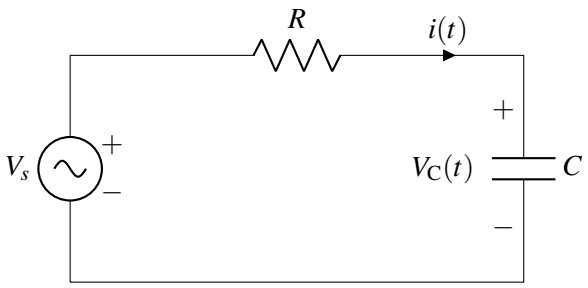
$$v(t) = \widetilde{V}e^{j\omega t} + \overline{\widetilde{V}}e^{-j\omega t}, \tag{1}$$

For a sinusoidal cosine wave $v(t) = V_0\cos(\omega t + \phi)$, a phasor is defined as:

$$\widetilde{V} = \frac{V_0}{2}e^{j\phi} \text{ and its complex conjugate } \overline{\widetilde{V}} = \frac{V_0}{2}e^{-j\phi}$$

$\widetilde{V}$ is a time-independent function, and is referred to as the phasor representation of $v(t)$.

The phasor analysis method consists of five steps. Consider the RC circuit below.



The voltage source is given by

$$V_s(t) = 8\cos\left(\omega t - \frac{\pi}{4}\right), \tag{2}$$

with $\omega = 10^3$ rad/s, $R = 1$ kΩ, and $C = 1$ μF.

Our goal is to obtain a solution for $V_C(t)$ and $i(t)$ with the sinusoidal voltage source $V_s(t)$.

(a) **Step 1: Convert $v(t)$ as a sum of complex exponentials**

All voltages and currents with known sinusoidal functions should be expressed in the standard exponential format. **Convert $v_s(t)$ into an exponential and write down its phasor representation $\widetilde{V}_s$.**

Solution:

$$v_s(t) = 8\cos\left(\omega t - \frac{\pi}{4}\right) \tag{3}$$

The two formulas below will help us convert a sinusoid into a sum of exponentials:

$$\cos(\theta) = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta} \text{ and } \sin(\theta) = \frac{1}{2j}e^{j\theta} - \frac{1}{2j}e^{-j\theta}$$

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Using the fact that $\cos(\theta) = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta}$,

$$v_s(t) = 8\left(\frac{1}{2}e^{j(\omega t - \frac{\pi}{4})} + \frac{1}{2}e^{-j(\omega t - \frac{\pi}{4})}\right) = 4e^{j(\omega t - \frac{\pi}{4})} + 4e^{-j(\omega t - \frac{\pi}{4})}$$
$$= 4e^{j\omega t} \cdot e^{-j\frac{\pi}{4}} + 4e^{-j\omega t}e^{j\frac{\pi}{4}}$$

The phasor is given by

$$\widetilde{V}_s = 4e^{-j\frac{\pi}{4}} \tag{4}$$

(b) **Step 2: Transform circuits to phasor domain**

The voltage source is represented by its phasor $\widetilde{V}_s$. The current $i(t)$ is related to its phasor counterpart $\widetilde{I}$. What are the phasor representations of R and C?

Solution:

$$Z_R = R \tag{5}$$

$$Z_C = \frac{1}{j\omega C} \tag{6}$$

(c) **Step 3: Cast KCL and/or KVL equations in phasor domain**

Use Kirchhoff's laws to write down a loop equation that relates all phasors in Step 2.

Solution: By KCL, the current through the resistor is equal to the current through the capacitor.

$$i(t) = i_r(t) = i_c(t) \rightarrow \widetilde{I} = \widetilde{I}_R = \widetilde{I}_C$$

By KVL, the sum of the voltages across the resistor and capacitor is equal to that of the source.

$$v_s(t) = v_c(t) + v_r(t) \rightarrow \widetilde{V}_s = \widetilde{V}_c + \widetilde{V}_R$$

Substituting the impedances in, we get:

$$\widetilde{V}_s = \widetilde{I}\frac{1}{j\omega C} + \widetilde{I}R = \widetilde{I}(R + \frac{1}{j\omega C})$$

(d) **Step 4: Solve for unknown variables**

Solve the equation you derived in Step 3 for $\widetilde{I}$ and $\widetilde{V}_c$. What is the polar form of $\widetilde{I}$ ($Ae^{i\theta}$, where A is a positive real number)?

Solution: We first solve for the current $\widetilde{I}$:

$$\widetilde{I} = \frac{\widetilde{V}_s}{R + \frac{1}{j\omega C}}$$

We plug in for $\omega = 10^3, R = 10^3\Omega, C = 10^{-6}\mu\text{F}$

$$\widetilde{I} = \frac{4e^{-j\frac{\pi}{4}}}{10^3 + \frac{1}{j10^3 10^{-6}}} = \frac{4e^{-j\frac{\pi}{4}}}{10^3 - 10^3 j} = \frac{4e^{-j\frac{\pi}{4}}}{10^3\sqrt{2}e^{-j\frac{\pi}{4}}} = \frac{2\sqrt{2}}{10^3}\ \text{A} = 2\sqrt{2}\ \text{mA}$$

To solve for $\widetilde{V}_c$ we use the voltage-current relationship $\widetilde{V} = \widetilde{I}\frac{1}{j\omega C}$

$$\widetilde{V}_c = \frac{1}{10^{-3}j}\frac{2\sqrt{2}}{10^3} = 2\sqrt{2}e^{-j\frac{\pi}{2}}\ \text{V}$$

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(e) **Step 5: Transform solutions back to time domain**

To return to time domain, we apply the fundamental relation between a sinusoidal function and its phasor counterpart. What is $i(t)$ and $v_c(t)$? What is the phase difference between $i(t)$ and $v_c(t)$?

Solution:

$$i(t) = Ie^{j\omega t} + \bar{I}e^{-j\omega t} = 4\sqrt{2}\cos(\omega t) \text{ mA} \tag{7}$$

$$v_C(t) = Ve^{j\omega t} + \bar{V}e^{-j\omega t} = 4\sqrt{2}\cos\left(\omega t - \frac{\pi}{2}\right) \text{ V} \tag{8}$$

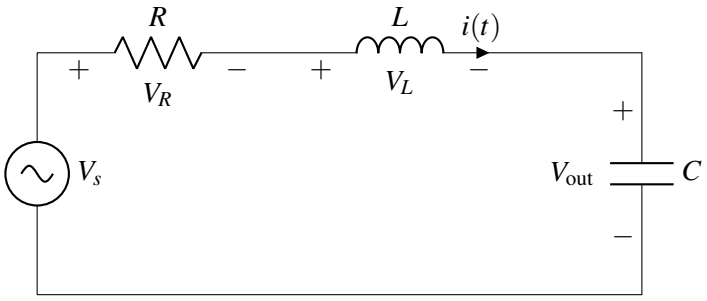
The phase difference between the two, with respect to $i(t)$ is $-\frac{\pi}{2}$

It's important to keep in mind that the phasor analysis above only applies to sinusoidal inputs. Don't make the mistake of doing phasor analysis on DC inputs!

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2. RLC circuit Phasor Analysis

In this question, we will take a look at an electrical systems described by second order differential equations and analyze it using the phasor domain. Consider the circuit below where $R = 3\text{k}\Omega$, $L = 1\text{mH}$, $C = 100\text{nF}$, and $V_s = 5\cos\left(1000t + \frac{\pi}{4}\right)$:



(a) What are the impedances of the resistor, inductor and capacitor, Z_R, Z_L, and Z_C?

Solution:

The impedance of a resistor is the same as its resistance

$$Z_R = 3000\Omega$$

We can find the frequency of the circuit by looking at V_s. The form of a cosine function is $A\cos(\omega t + \phi)$ where A is amplitude, ω is frequency, and ϕ is phase. In this case, the frequency is $1000\frac{\text{rad}}{\text{s}}$

$$Z_L = j\omega L = j1000 * 10^{-3} = j1\Omega$$
$$Z_C = \frac{1}{j\omega C} = \frac{1}{j1000 * 10^{-7}} = -j10^4\Omega$$

(b) Solve for $\widetilde{V}_{\text{out}}$ in phasor form.

Solution:

converting V_s into phasor form, we have

$$\widetilde{V}_s = \frac{1}{2}|A|e^{j\phi} = \frac{5}{2}e^{j\frac{\pi}{4}}$$

The circuit given is a voltage divider. Since impedances act like resistors, we can use the same equation as the resistive voltage divider.

$$\widetilde{V}_{\text{out}} = \widetilde{V}_s\frac{Z_C}{Z_R + Z_L + Z_C} = \widetilde{V}_s * \left|\frac{Z_C}{Z_R + Z_L + Z_C}\right| e^{j*\angle\left(\frac{Z_C}{Z_R+Z_L+Z_C}\right)} \tag{1}$$

We can solve for the magnitude and angle of the divider using

$$\left|\frac{Z_C}{Z_R + Z_L + Z_C}\right| = \frac{|Z_C|}{|Z_R + Z_L + Z_C|} = \frac{10^4}{\sqrt{3000^2 + (1 - 10^4)^2}} = 0.958$$
$$\angle\left(\frac{Z_C}{Z_R + Z_L + Z_C}\right) = \angle(Z_C) - \angle(Z_R + Z_L + Z_C) = \text{atan2}\left(\frac{\Im\text{m}(Z_C)}{\Re\text{e}(Z_C)}\right) - \text{atan2}\left(\frac{\Im\text{m}(Z_R + Z_L + Z_C)}{\Re\text{e}(Z_R + Z_L + Z_C)}\right)$$
$$\angle\left(\frac{Z_C}{Z_R + Z_L + Z_C}\right) = -0.2915 \text{ rad}$$

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Plugging back into (1)

$$\widetilde{V}_{\text{out}} = \frac{5}{2} e^{j\frac{\pi}{4}} * 0.958 e^{-j0.2915} = 2.395 e^{j0.494}$$

(c) What is V_{out} in the time domain?

Solution:

$$V_{\text{out}}(t) = 4.79 \cos(1000t + 0.494) \text{ V}$$

(d) Solve for the current $i(t)$

Solution:

$$\widetilde{i} = \frac{\widetilde{V}_{\text{s}}}{Z_R + Z_L + Z_C} = \frac{|\widetilde{V}_{\text{s}}|}{|Z_R + Z_L + Z_C|} e^{j\left(\angle \widetilde{V}_{\text{s}} - \angle (Z_R + Z_L + Z_C)\right)} = 2.395 * 10^{-4} e^{j2.0647}$$

Going back to the time domain:

$$i(t) = 4.790 * 10^{-4} \cos(1000t + 2.0647) \text{ A}$$

(e) Solve for the transfer function $H(\omega) = \frac{\widetilde{V}_{\text{out}}}{\widetilde{V}_{\text{s}}}$

Leave your answer in terms of R, L, C, and ω.

Solution:

Looking back at equation (1)

$$\widetilde{V}_{\text{out}} = \widetilde{V}_{\text{s}} \frac{Z_C}{Z_R + Z_L + Z_C}$$

Rearranging we get

$$H(\omega) = \frac{\widetilde{V}_{\text{out}}}{\widetilde{V}_{\text{s}}} = \frac{Z_C}{Z_R + Z_L + Z_C} = \frac{\frac{1}{j\omega C}}{R + j\omega L + \frac{1}{j\omega C}}$$

$$H(\omega) = \frac{1}{LC(j\omega)^2 + jRC\omega + 1}$$

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3. RC Circuit

Consider the circuit below, assume that when $t \leq 0$, the capacitor has no charge stored ($V_c(t=0)=0$). At $t=0$, the switch closes. Assume that $V_s = 5\,\text{V}$, $R = 100\,\Omega$, and $C = 10\,\mu\text{F}$.

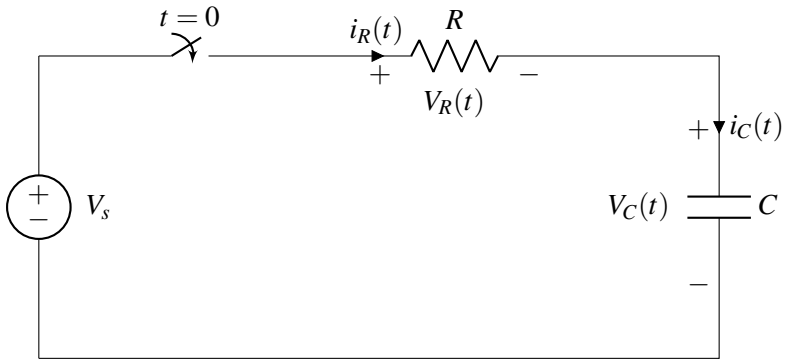


Figure 1: RC Circuit with Voltage Source

(a) Write out the KCL equations associated with the circuit when the switch is closed.
Solution:

We denote the voltage to the left of resistor to be V_s.

By KCL, the current through the resistor

$$i_R = i_C.$$

Ohm's law tells us that

$$i_R = \frac{V_s - V_c}{R}$$

The voltage-current relationship of a capacitor tells us that

$$i_C = C\frac{dV_c}{dt}$$

Therefore,

$$\frac{V_s - V_c}{R} = C\frac{dV_c}{dt}$$

(b) Write out the differential equation for $V_c(t)$ after the switch closes.
Solution:

From the previous problem we know that when the switch is closed,

$$\frac{V_s - V_c}{R} = C\frac{dV_c}{dt}$$

Thus we obtain

$$\frac{dV_c}{dt} + \frac{V_c}{RC} = \frac{V_S}{RC}$$

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(c) What is the initial condition for $V_c(t)$ (i.e. $V_c(t=0)$) and what is $V_c(t \to \infty)$?

Solution: No charge is on the capacitor before time $t = 0$. Using $q = VC$, we know that $V_c = 0\,\text{V}$ before $t = 0$. At $t = 0$, the switch closes. Since voltage across the capacitor cannot change instantaneously, $V_c(t=0) = 0$. As t goes to infinity, the capacitor will become fully charged and the current goes to zero. Therefore, the voltage of the capacitor equals the voltage source: $V_c(t \to \infty) = V_S$.

(d) Using the initial conditions found in the previous parts, find an expression for $V_c(t)$ in terms of V_s, R, and C.

Solution: To find an expression for $V_{out}(t)$:

$$V_c(t) = V_{out}(t)$$

$$\frac{dV_{out}(t)}{dt} + \frac{1}{RC}V_{out}(t) = \frac{V_S}{RC}$$

Using substitution of variables:

$$x(t) = V_{out}(t) - V_S$$

$$V_{out}(t) = x(t) + V_S$$

$$\frac{dV_{out}(t)}{dt} = \frac{dx(t)}{dt}$$

$$\frac{dx(t)}{dt} + \frac{1}{RC}x(t) = 0$$

$$x(t) = Ae^{-\frac{t}{RC}}$$

Now substituting back to find V_{out}:

$$V_{out}(t) = V_S + Ae^{-\frac{t}{RC}}$$

Using our initial condition:

$$V_{out}(0) = 0 = V_S + A$$

$$A = -V_S$$

$$V_{out}(t) = V_S\left(1 - e^{-\frac{t}{RC}}\right)$$

$V_c(t) = V_S(1 - e^{-\frac{t}{\tau}})$ where $\tau = RC$ (τ explained in next part)

Aside: τ is the time constant that we denote for RC circuits. Its definition is the time required to charge the capacitor from an initial voltage of zero to about 63.2% of the value of V_s, or to discharge the capacitor to about 36.8% of its initial charge voltage. For an RC circuit it is equal to $\frac{1}{\alpha}$ where α is the eigenvalue of the RC circuit.

(e) OPTIONAL: On what order of magnitude of time (nanoseconds, milliseconds, 10's of seconds, etc.) does this circuit settle (V_c is $> 95\%$ of its value as $t \to \infty$)?

Solution:

The time constant τ of an RC circuit is just $\tau = RC$. For our circuit:

$$\tau = RC = 100\,\Omega \cdot 10\,\mu\text{F} = 0.001\,\text{s}$$

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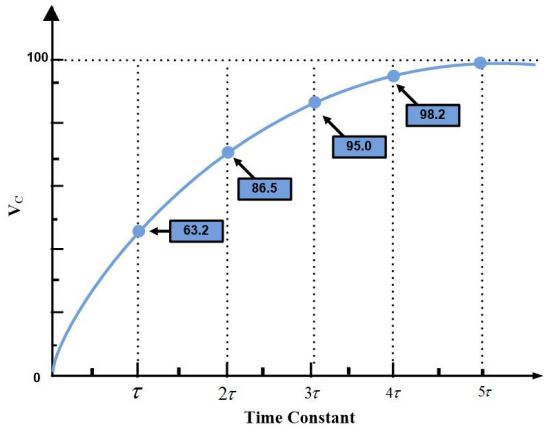


Figure 2: Different values of capacitor voltage at different times, relative to τ.

After 3 time constants, the voltage will be 95% of its steady state value

$$3\tau = 0.003\,\text{s}$$

The circuit will settle on the order of milliseconds. Alternatively, this value can be found by using algebra:

$$0.95V_S = V_S(1 - e^{-\frac{t}{RC}})$$
$$-0.05 = -e^{-\frac{t}{RC}}$$
$$0.05 = e^{-\frac{t}{RC}}$$
$$ln(0.05) = -\frac{t}{RC}$$
$$-3 = -\frac{t}{0.001}$$
$$t = 0.003\,\text{seconds}$$

(f) OPTIONAL: Give 2 ways to reduce the settling time of the circuit if we are allowed to change one component in the circuit.

Solution:

To reduce settling time, reduce τ. We can achieve this by

i. Lowering the value of R or
ii. Lowering the value of C.

Notice how the value of V_s does not change the settling time.

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